

Questions for 2 way ANOVA

1) Employee Training — 4 programs $\times$ 5 departments (blocks)

Four training programs (levels) evaluated across 5 departments (blocks). Outcome: % productivity improvement. At 0.01 level of significance conclude that average productivity is same under department & Programs

Program\Dept	Dept 1	Dept 2	Dept 3	Dept 4	Dept 5
1 (Workshop)	12	15	11	14	13
2 (Online)	10	12	9	11	10
3 (Hybrid)	14	17	13	16	15
4 (Coaching)	16	18	15	17	16

2) Customer Script Evaluation — 3 scripts $\times$ 8 teams (blocks)

Three call scripts tested across 8 teams. Outcome: customer satisfaction (0–100).

At 0.05 level of significance conclude that average customer satisfactions is same under team.

Script\Team	T1	T2	T3	T4	T5	T6	T7	T8
1	78	80	83	76	81	79	77	82
2	86	84	88	85	87	86	83	89
3	74	76	78	73	75	77	72	79

3) Packaging Design — 2 designs $\times$ 6 regions (blocks)

A simpler test: two packaging alternatives across 6 regions. Outcome: mean weekly sales (\$1000s).

At 0.1 level of significance conclude that average weekly sale is same under regions

Design\Region	R1	R2	R3	R4	R5	R6
A	55	50	58	62	49	53

Design\Region	R1	R2	R3	R4	R5	R6
B	61	56	63	67	54	59

4) Pricing Strategy — 4 strategies $\times$ 6 stores (blocks)

Four pricing strategies tested across 6 stores. Outcome: weekly units sold.

At 0.01 level of significance conclude that average weekly units sold is same under strategies

Strategy\Store	S1	S2	S3	S4	S5	S6
1 (High)	150	162	158	151	155	160
2 (Promo)	170	175	172	168	173	176
3 (Bundle)	160	166	164	161	165	167
4 (Penetration)	180	185	182	179	183	186

5) Interface Layout — 3 layouts $\times$ 7 employee groups (blocks)

Three dashboard layouts across 7 groups. Outcome: task completion time (seconds).

At 0.01 level of significance conclude that average completion is different under dashboards

Layout\Group	G1	G2	G3	G4	G5	G6	G7
1	342	330	355	348	339	347	351
2	298	290	305	302	295	300	303
3	318	310	325	320	315	322	327

6) Advertising Channels — 5 channels $\times$ 6 product categories (blocks)

Five ad channels tested across 6 product categories. Outcome: monthly online traffic increase (%). At 0.01 level of significance conclude that average monthly online traffic is same under categories & Channels

Channel\Category	C1	C2	C3	C4	C5	C6
1 (Social)	12	14	13	11	12	15
2 (Search)	20	22	21	19	20	23

Channel\Category	C1	C2	C3	C4	C5	C6
3 (Email)	8	9	10	7	8	11
4 (Display)	15	16	14	13	15	17
5 (Affiliate)	10	11	12	9	10	13

7) A company is evaluating the effectiveness of five different business growth strategies to determine which one leads to the highest average revenue increase. The strategies being tested are:

1. Digital Marketing Expansion
2. Strategic Partnerships
3. Product Diversification
4. Market Penetration Pricing
5. Customer Loyalty Programs

The company implements each strategy in five different regions, where the regions act as blocks in a randomized complete block design. After a set period, the revenue increase (in thousands of dollars) for each region is recorded as follows:

Strategy	Region 1	Region 2	Region 3	Region 4	Region 5
Digital Marketing Expansion	15	18	22	17	20
Strategic Partnerships	25	30	28	27	29
Product Diversification	12	14	15	13	16
Market Penetration Pricing	20	19	21	22	18
Customer Loyalty Programs	10	12	11	13	14

At the 5% significance level, perform a randomized complete block design (RCBD) analysis of variance to determine whether there is a statistically significant difference in the mean revenue increase among the five business growth strategies, after accounting for regional effects.

8) A financial analyst at a bank is conducting an experiment to determine whether there are significant differences in the returns of five different investment portfolios across four different market conditions. The portfolios consist of different asset allocations, and the market conditions include bullish, bearish, stable, and volatile markets.

The returns (in percentage) for each portfolio in each market condition over the past year are recorded as follows:

Portfolio	Bullish	Bearish	Stable	Volatile
Portfolio 1	8.5	-2.3	4.0	1.2
Portfolio 2	9.1	-3.0	3.5	2.0
Portfolio 3	7.8	-1.8	4.5	1.5
Portfolio 4	10.0	-4.2	5.0	3.0
Portfolio 5	6.5	-2.0	3.0	2.5

Using a significance level of $\alpha = 0.05$, determine whether there are significant differences in the returns of the investment portfolios across the different market conditions.